12.8 Infections of the Nervous System

Some of the most devastating infectious diseases are those that affect the nervous system, especially the brain and the spinal cord. Damage to these areas can lead to deafness, blindness, learning disabilities, paralysis and death. Microbial infections of CNS are infrequent but often have serious consequences. In pre antibiotic times, they were almost always fatal. An infection of CNS can be life threatening condition, especially for children with weakened immune system. These infections need quick diagnosis and immediate treatment by an infectious disease specialist. Bacteria, Fungi and viruses are the most common causes of CNS infections.

12.8.1 Structure of Nervous System

The human nervous system is organized into two divisions: The Central Nervous System (CNS) and Peripheral Nervous System (PNS). The Central Nervous System (CNS) consists of brain and spinal cord. It controls most functions of the body and mind. The peripheral nervous system (PNS) consists of all the nerves that branch off from the brain and spinal cord. These peripheral nerves are the lines of communication between the CNS, the various parts of the body and the external environment (Figure 12.19).

Brain and spinal cord are covered by three layers of membranes called meninges. These layers are the outermost dura mater, the middle arachnoid mater, and the innermost pia mater. Between the pia mater and arachnoid membranes is a space called the subarachnoid space, in which there is cerebrospinal fluid (CSF) circulating.

12.8.2 Barriers of CNS

Dyes such as Trypan blue injected into the systemic circulation stain virtually all tissues, with the exception of the brain and spinal cord. This blood brain

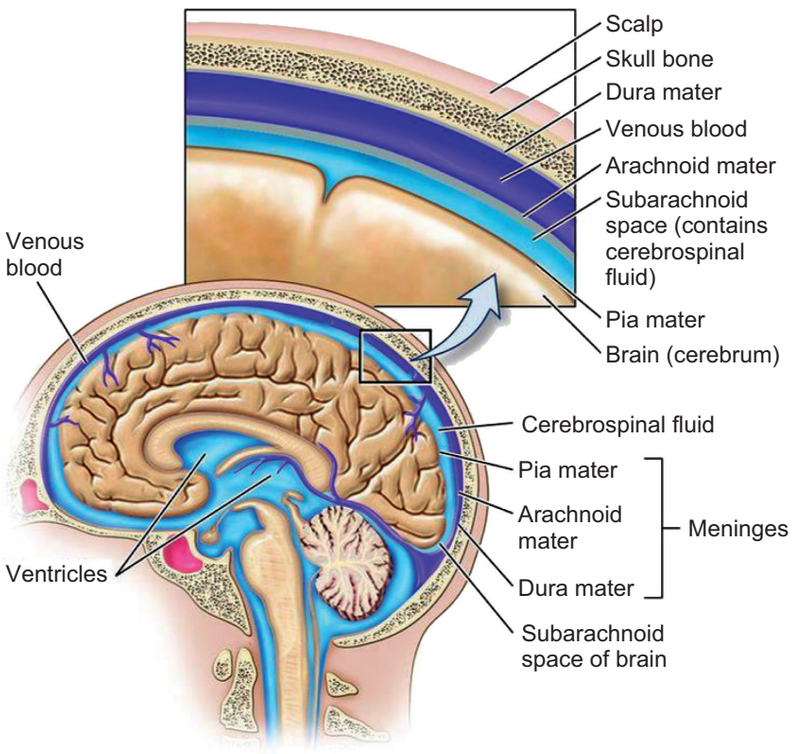


Figure 12.19: Structure of central nervous system

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